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26 AUGUST 1958

SUPERSEDING
MIL-C-10338A
29 DECEMBER 1950

MILITARY SPECIFICATION

CHEMICAL AGENT, CHLOROACETOPHENONE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one grade of chloroacetophenone ($C_6H_5COCH_2Cl$).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

RR-D-741 — Drums, Steel, Type 6C (for Inflammable Solids or Oxidizing Materials).

MILITARY

MIL-P-3803 — Plastic, Polyethylene, Molded and Extruded Shapes, Sheets, and Tubing.

STANDARD

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Copies may be obtained from Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

INTERSTATE COMMERCE COMMISSION

49 CFR 71-78 — Interstate Commerce Commission Rules

FSC 1365

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and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (*1949 Edition—Revised 1950*) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71–78 (Rev. 1950).")

3. REQUIREMENTS

3.1 Form. The chloroacetophenone shall be in the form of free-flowing crystals.

3.2 Chemical. The material shall conform to the chemical requirements of table I.

TABLE I. *Chemical requirements*

Requirement	Percent	Test method
Purity	95.0 minimum	4.3.1
Moisture	0.35 maximum	4.3.2
Acidity (as HCl)	0.1 maximum	4.3.3

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Unless otherwise specified herein, the contractor is responsible for the examination and testing of the chloroacetophenone prior to submission for acceptance.

4.2 Inspection provisions.

4.2.1 *Lotting.* A lot shall consist of the material produced by one manufacturer in not more than 24 consecutive hours under essentially the same manufacturing conditions and with no change in materials, provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot.

4.2.1.1 *Batch.* A batch is defined as that quantity of material which has been manufactured by some unit chemical process and/or subjected to some physical mixing operation intended to make the final product substantially uniform.

4.2.2 *Sampling.*

4.2.2.1 *For examination.* Sampling shall be conducted in accordance with Standard MIL–STD–105.

4.2.2.2 *For test.* A sample shall be taken from each lot, the size to be calculated on the basis of one-tenth of the square root of the number of containers in the lot raised to the next highest whole number. If there are fewer than three containers in a lot, each container shall be sampled. In all other cases, no fewer than three containers shall be selected. From each container in the sample a representative one pound specimen shall be taken and placed in separate, clean, dry containers which are labeled to identify the lot and container represented.

4.2.3 *Examination.*

4.2.3.1 *Nondestructive.* Each lot shall be inspected in accordance with the classification of defects and Standard MIL–STD–105.

4.2.3.2 *Test.* Equal portions of each specimen selected in accordance with 4.2.2.2 shall be thoroughly mixed to form a composite specimen. The composite specimen shall be subjected to the test specified in 4.3.

4.2.3.3 *Form.* The composite sample shall be visually examined for compliance with 3.1.

4.2.3.4 *Classification of defects.*

4.2.3.4.1 *Preparation for delivery.* (Level A only.)

Categories and defects

Major—AQL 4.0 percent defective

1. Container damaged.
2. Container leakage.
3. Container not as specified.
4. Container liner missing or not as specified.
5. Container closure incorrect.
6. Marking incorrect, missing or illegible.

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4.3 Test. Distilled water and analytical reagent grade chemicals shall be used throughout the test. Where applicable a blank determination shall be run and corrections applied where significant.

4.3.1 *Purity.* For the distillation apparatus refer to figure 1. Weigh the distillation flask A, including the capillary assembly B, to the nearest 0.01 gram (gm.). Transfer approximately 200 gm. of the specimen to the flask and again weigh to the nearest 0.01 gm. Weigh, to the nearest 0.01 gm., the receiving flask D and the condenser C with the two rubber stoppers attached. Connect the apparatus as shown on figure 1. Immerse the bulb of A in an oil bath containing a thermometer. Immerse D in an ice bath, keeping its exit arm well covered with ice. Apply vacuum and adjust to a residual pressure of 15 ± 2 millimeters (mm.) of mercury. After the correct pressure has been obtained it shall remain constant for at least 5 minutes as an indication of the absence of leakage. Apply heat to the oil bath and melt the chloroacetophenone. Adjust the stopcock on B to admit just sufficient air to prevent bumping. Raise the temperature very slowly as the boiling point is approached (approximately 135°C.) and distill at a rate sufficiently low to avoid bumping or surging over the chloroacetophenone into C. Melt any chloroacetophenone that freezes in C by flicking the flame of a burner against the tube. Continue until the distillation ceases and the chloroacetophenone on the interior walls of A has been removed by the small amount of air admitted through B. Admit air into the apparatus and remove the oil and ice baths. Clean and dry the exterior of the tared apparatus. Weigh the assembly of A and B, the assembly of C and stoppers, and the receiver D to the nearest 0.01 gm. Calculate the percent of chloroacetophenone recovered as follows:

$$\text{Percent chloroacetophenone recovered} = A = \frac{100 \times (E + F)}{W}$$

where:

E = increase in weight of receiver D, in gm.

F = increase in weight of condenser assembly C and stopper, in gm.

W = weight of specimen used, in gm.

Since the material remaining in C may have a different purity than that of the main distillate, combine the two by melting the chloroacetophenone in D by gentle application of heat. Warm C and pour the melted chloroacetophenone from D through C into a 250-ml. beaker. Thoroughly mix the completely melted mass by stirring. Weigh to the nearest 0.1 milligram (mg.) a blown glass bulb having a diameter of approximately 1.5 centimeter and a capillary tip, introduce the tip into the melted chloroacetophenone, and allow about 0.2 gm. to be drawn into the bulb. Wipe the capillary, seal the tip in the flame, cool, and weigh to the nearest 0.1 mg. Transfer to a Parr sulphur bomb cup. Add 0.3 gm. of 40-mesh sucrose and 15 gm. of calorific-grade sodium peroxide. Assemble and tighten the bomb, shake it thoroughly, and place it in a holder within a shield (see 6.4). Ignite the contents of the bomb by directing the flame of an oxy-gas burner against the bottom of the cup. After the initial ignition, continue the heating until the lower half of the cup attains an even cherry-red color. Remove the bomb from the shield, allow to stand for about five minutes and then cool in water. Remove the outer fastenings without disturbing the cap, wash the exterior of the cup and cap with water and discard the washings. Remove the cap and place the cup on its side in a 600-ml. beaker containing about 100 ml. of water previously heated

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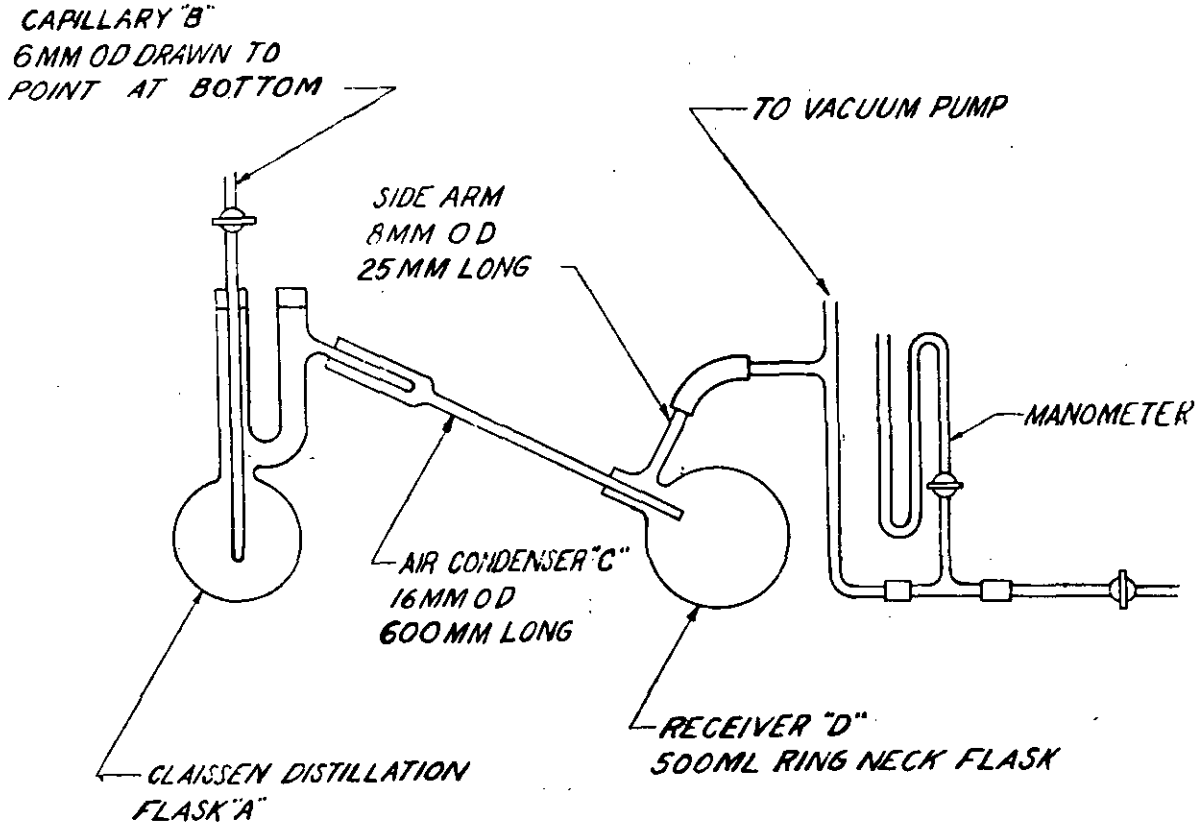


FIGURE 1. *Apparatus for vacuum distillation of chloracetophenone.*

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almost to boiling. Wash with a fine jet of water, catching the washings in the beaker. With a pair of tongs, lay the fusion cup on its side in the same beaker of hot water and cover the beaker immediately with a watch glass to act as a spattershield. After the melt has dissolved, remove the cup and wash thoroughly with a fine jet of hot water. Retain all the washings in the beaker. Add a 1 to 1 solution of nitric acid in small portions, allowing sufficient time between additions of the acid for the completion of the reaction, so as to avoid a large excess of acid. When the solution is clear, allow to cool, and determine the total chlorine by the Volhard method. Calculate the percent purity of recovered chloroacetophenone (P) as follows:

$$P = \frac{15.46 \times (VN - BM)}{G}$$

where:

V = volume of silver nitrate used in titration, in ml.

N = normality of silver nitrate.

B = volume of potassium thiocyanate solution used, in ml.

M = normality of potassium thiocyanate solution.

G = weight of specimen of recovered chloroacetophenone, in gm.

Calculate the percent purity of the specimen as follows:

$$\text{Percent purity} = \frac{A \times P}{100}$$

where:

A = percent of chloroacetophenone recovered.

P = percent purity of recovered chloroacetophenone.

4.3.2 *Moisture.* Use the apparatus shown in figure 2. Weigh approximately 25 gm. of the specimen in flask F. Raise the graduate D until the saturated salt solution in gas washing bottle C rises to the top of the stopper, and close pinchcock G. Now quickly insert the stopper tightly in F, open G, raise D until the solution in D is level with the solution in C, close G, raise the tube in D to the surface of the salt solution, and read the number of ml. in the graduate. Remove stopper from F and add 5 gm. of calcium carbide, which has been ground to pass a 5-mesh and be retained on an 80-mesh sieve. Quickly replace the stopper in F, open G, and melt the chloroacetophenone by immersing F in a bath of hot water. Shake the flask vigorously until the flow of salt solution from C to D ceases, allow the flask to cool to room temperature, raise D until the solution in D is level with the solution in C; close G, raise the tube in D, to the surface of the salt solution, and read the number of ml. in the graduate. Take barometer and room temperature readings. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{0.0578 \times P \times V}{(273 + T) \times W}$$

where:

P = barometer reading in mm. of mercury.

V = volume of acetylene, in ml.

T = temperature in degrees C.

W = weight of specimen, in gm.

4.3.3 *Acidity.* Weigh approximately 10 gm. of the specimen in a 600-ml. beaker, add 300 ml. of cold water; stir thoroughly and break up the chloroacetophenone as fine as possible. Allow to stand for 15 minutes, stirring occasionally, and then filter. Titrate the filtrate with 0.1 N sodium hydroxide, using phenolphthalein as indicator. Add to the chloroacetophenone residue 300 ml. of water

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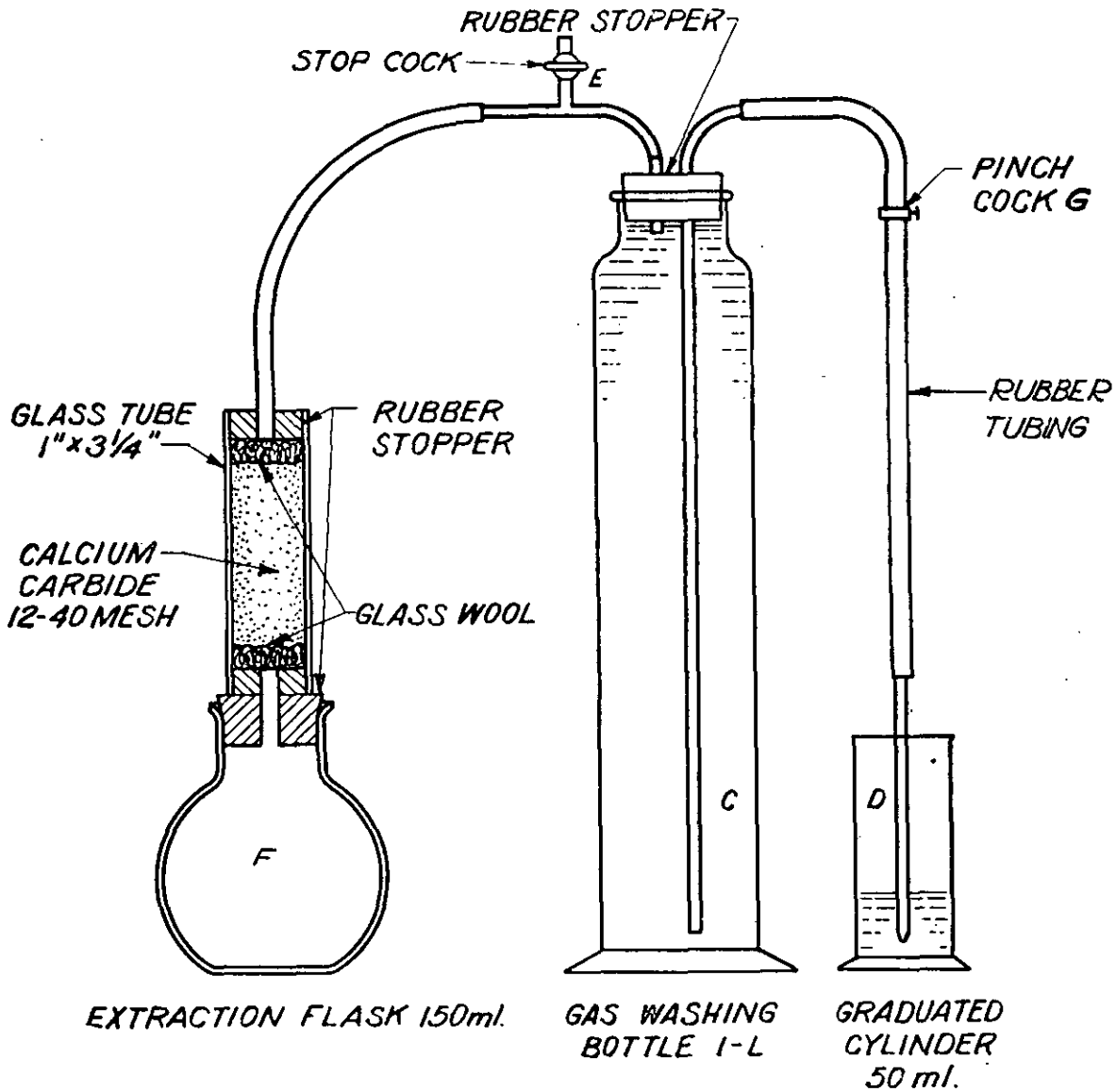


FIGURE 2. *Apparatus for the determination of moisture by the generation of acetylene.*

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and allow to stand 15 minutes. Then filter and titrate the filtrate with 0.1 N sodium hydroxide. Ascertain the total volume of sodium hydroxide used in the two titrations and calculate the acidity as follows:

$$\text{Percent acidity as hydrochloric acid} = \frac{3.65 \times V \times N}{W}$$

where:

V = total volume of sodium used in the titration, in ml.

N = normality of the sodium hydroxide.

W = weight of specimen used, in gm.

4.4 Rejection criteria. If composite specimen selected as speciled in 4.2.3.2 fails to conform to this specification, the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A.* The chloroacetophenone shall be packed in 16 gauge, 55 gallon capacity, straight sided, fully removable head drums. The drums shall conform to either types III or IV of Specification RR–D–741 or ICC specifications 5 or 5B. The removable heads of the drums shall be provided with bolt type closure rings with insert type gaskets fabricated from a material which shall not affect nor be affected by the contents. The drums shall be provided with a minimum thickness of 0.0030 polyethylene liner conforming to grade 1 material of Specification MIL–P–3803. The liner shall be constructed in such manner as to assure that the chloroacetophenone shall not come in contact with the interior surfaces of the drum or cover before, during, or after filling and subsequent closure.

5.1.2 *Level B.* Not applicable.

5.1.3 *Level C.* The chloroacetophenone shall be packed in accordance with manufacturer's commercial practice to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with ICC Regulations, Uniform Freight Classification Rules or any other common carrier regulations applicable to the mode of transportation.

5.2 Marking. In addition to any special markings required by the contract or order, all containers shall be marked in accordance with Standard MIL–STD–129.

5.2.1 *Precautionary marking.* Each individual container shall be marked as follows:

Caution

Avoid breathing dust or vapor (military type canister protective mask required).

Avoid contact with eyes, skin and clothing.

6. NOTES

6.1 Intended use. The chloroacetophenone covered by this specification is intended for use in munitions as a tear gas.

6.2 Ordering data. Procurement documents should specify the following:

(a) The title, number and date of this specification.

(b) The level of packing required.

6.3 Safety precautions. Safety precautions governing the handling of chloroacetophenone should be observed by personnel handling this material in order to avoid injury. This material will excite lachrymation.

6.4 Parr bombs should be enclosed by a suitable shield to protect the operator from injury if the bomb explodes accidentally.

Notice. When Government drawings, specifications, or other data are used for any purpose other than

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in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person of corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Chemical Corps
Navy—Bureau of Ordnance
Air Force

Other interest:
Army—M

Preparing activity:
Army—Chemical Corps

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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